
When Shift Happens - Confounding Is to Blame

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Abstract

1 Distribution shifts inject uncertainty that erodes the robustness and generalization
2 of machine learning models. Prevailing wisdom holds that learning causal-invariant
3 representations is the route to robustness under such shifts, yet a growing body of
4 empirical evidence points in the opposite direction: (i) empirical risk minimiza-
5 tion (ERM) matches, and frequently surpasses, state-of-the-art out-of-distribution
6 (OOD) generalization methods, and (ii) its OOD performance improves precisely
7 when *all* available covariates—not only the causal ones—are exploited. We resolve
8 this apparent paradox by identifying hidden confounding as its common cause.
9 Shifts in hidden confounders propagate into the observable data distribution in
10 ways that systematically violate the invariance assumptions on which existing
11 OOD methods rely. Under these conditions we prove that effective generalization
12 demands learning environment-specific relationships: invariant representations,
13 though necessary, are provably insufficient. We further establish that augmenting a
14 model with proxies for the hidden confounders directly counteracts the difficulty
15 introduced by confounding shift. Together, these results supply both a theoretical
16 account of why ERM is so effective under real-world shift and concrete, action-
17 able guidance for designing robust OOD generalization algorithms and principled
18 covariate-selection strategies.

1 Introduction

19 Generalization—the capacity to draw reliable conclusions about unseen data from observed data—is a
20 foundational concern across the sciences. It recurs, under different names, wherever conclusions must
21 transfer beyond the conditions under which they were obtained: in medicine and the social sciences
22 as *external validity*, the requirement that findings from one population carry over to another [13]; in
23 ecology as *space-for-time substitution*, in which spatial variation stands in as a proxy for temporal
24 change to infer long-term ecological trends [60]; and in engineering as *robust control*, in which
25 a model must sustain its performance in the presence of unmodeled disturbances [35]. Machine
26 learning has become the principal computational instrument for learning models that generalize in
27 this sense [100, 88, 49].

29 Generalization in machine learning hinges on robustness to the distribution shift that separates training
30 data from test data. A model trained to predict *food stamp reciprocity* from *household attributes* in
31 one state, for instance, should retain its predictive accuracy when deployed in another. We say a
32 model achieves *out-of-distribution (OOD) generalization* when it performs well simultaneously on
33 in-distribution (ID) data—drawn from the training distribution—and on out-of-distribution (OOD)
34 test data. Over the past decade, several distinct families of methods have been developed to secure
35 provable OOD generalization under specific structural assumptions: domain generalization [56,
36 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94].
37 Strikingly, however, once model selection is performed carefully [30, 38, 87], standard *empirical risk*
38 *minimization (ERM)* [86] attains OOD performance that is competitive with, and often superior to,
39 these specialized methods across a broad spectrum of real-world applications [30, 24, 50, 57, 67, 87].
40 A recent study of 16 real-world tabular datasets [57] sharpens the puzzle further: incorporating *all*
41 available covariates when predicting the outcome—irrespective of whether they directly influence
42 it—systematically improves OOD generalization. These observations run counter to the field’s

prevailing assumptions and demand a principled account of the mechanisms that actually govern OOD generalization.

Distribution shift arises characteristically when data are collected across heterogeneous environments [71]. *Food stamp reciprocity*, for example, varies from state to state because each state applies distinct eligibility rules, giving rise to environment-specific distributions. Within such data, some statistical relationships between covariates and outcome persist unchanged across every environment; these are the *invariant relationships* [6], and identifying and learning them is the canonical route to OOD generalization [6, 59, 31, 66, 56, 98, 63]. A particularly important instance is *causal invariance*, in which the causal mechanisms themselves remain invariant across environments [71, 59, 6]. In practice, however, not every covariate relevant to predicting the outcome is observed—data-collection limits, privacy constraints, and measurement error routinely leave variables unmeasured [14, 52, 20, 48]—and this incompleteness disrupts the very invariances on which generalization is predicated. The difficulty becomes acute when the unobserved variables are *confounders* that jointly drive the observed covariates and the outcome. Such hidden confounders are pervasive, and shifts in their distributions delineate distinct environments; disregarding these shifts not only degrades generalization [40, 4, 85, 62] but actively induces the model to learn the *wrong* covariate–outcome relationships.

Despite its centrality, the effect of hidden confounding shift on OOD generalization has received little systematic attention [4, 40, 85, 62]. This work closes that gap, providing a unified theoretical and empirical account that explains the recent, seemingly anomalous findings in the literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- **A causal formulation of confounding shift.** We formalize, from a causal standpoint, why hidden confounding shift must be treated as a first-class phenomenon in OOD generalization, and we explain precisely why enlarging the covariate set with informative, non-causal variables improves performance (§ 3).
- **Invariance is provably insufficient.** We prove that maximizing the predictive information between model predictions and true outcomes *requires* learning environment-specific relationships; invariant representations, while necessary, are demonstrably insufficient on their own (§ 4.1, § 4.2).
- **Informative covariates as confounder proxies.** We show that any covariate informative of the outcome or of the hidden confounders increases the predictive information between predictions and true outcomes, establishing principled covariate selection as an essential lever under hidden confounding shift (§ 4.3).
- **Empirical corroboration across real and synthetic data.** Experiments on eight real-world tabular benchmarks and controlled synthetic data confirm that (i) hidden confounding shift is pervasive in real-world tabular data, (ii) the capacity to learn environment-specific relationships is positively correlated with both ID and OOD test accuracy, and (iii) incorporating informative, non-causal covariates improves generalization (§ 5).

Taken together, our results not only explain the surprising strength of ERM under real-world distribution shift but also expose the subtle yet unavoidable role that hidden confounding plays in OOD generalization—and, in doing so, chart the requirements for methods that confront it directly.

2 Related work

Out-of-distribution (OOD) generalization spans several closely related problem settings, among them domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These settings differ primarily in the information they assume at training and test time: domain adaptation presumes access to unlabeled test data; domain generalization relies on environment labels during training; federated learning operates within a collaborative framework subject to communication-efficiency and privacy constraints when data span multiple environments; and OOD detection targets the identification of samples that deviate markedly from the training distribution. What unites the majority of these approaches is a single organizing objective—learning invariant relationships. A growing line of work, however, argues that invariance alone is inadequate and that additional inductive biases are indispensable for strong generalization [46, 72, 93]. In particular, Ye et al. [93] establish that feature invariance is necessary but not sufficient, foregrounding the complementary role of feature *informativeness*. Our analysis places invariance and informativeness on an equal footing and characterizes how the two jointly determine OOD generalization.

Proxy-variable adjustment. When confounders are observed during training but unobserved at test time, Landeiro and Culotta [40] adjust for the induced confounding shift to restore classifier performance. Building on proxy-based adjustment for causal-effect estimation [55, 39], a recent strand of domain adaptation recasts unknown distribution shift as a problem of causal-effect identification [4, 85], and OOD generalization under hidden confounding shift has been studied under an assumption of overlapping confounder support [62]. Whereas these methods prescribe *how* to adjust, our contribution is to explain *why* proxy variables enhance OOD generalization in the first place, grounding their effectiveness in an information-theoretic decomposition of predictive information.

All-variable models versus causal models. Nastl and Hardt [57] introduced a benchmark that partitions covariates into four categories—causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates—and demonstrated that, across all 16 datasets, models trained on the full covariate set Pareto-dominate those restricted to causal or arguably causal subsets on both ID and OOD data. This empirical regularity has, until now, wanted a theoretical explanation; we supply one, deriving the scenarios and mechanisms under which it must hold. For linear causal models, anchor regression [68] interpolates between the two estimation paradigms—using all observed covariates versus using only causal covariates—but our object is different: we explain the effect of adding non-causal covariates specifically under hidden confounding shift. Eastwood et al. [21] show that unstable covariates can boost performance when they carry label information and are conditionally independent of the stable covariates given the label, adjusting for the shift by inspecting the unlabeled test domain; yet on a medical dataset not purpose-built for this setting [8], ERM remains competitive with their method, consistent with Nastl and Hardt [57]. This aligns with the broader result that, under well-specified covariate shift, maximum likelihood estimation attains minimax optimality for OOD generalization [26]. The decisive point is that real-world settings are seldom well-specified, precisely because of hidden confounding shift—the phenomenon this work places at its center.

3 Manifestations of hidden confounding shift

This section establishes the background on hidden confounding shift and motivates the necessity of addressing it directly, using an example built around the identification of the causal effect of observed covariates on the outcome.

Types of distribution shift. For covariates $\mathbf{X}$ and target Y , two environments e and e' may differ through any of the several distribution shifts catalogued in Table 1. Each of these shifts is, at root, a downstream manifestation of a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes $\mathbf{X}$, Y , or both (Figure 1). Because the observed distributions are marginals and conditionals of the joint over $(U, \mathbf{X}, Y)$, a shift in $\mathbb{P}(U)$ propagates mechanically into the distributions over $\mathbf{X}$ and Y alone. Writing $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U) \mathbb{P}(\mathbf{X} | U)$ makes this explicit: whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, it follows that $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The identical argument accounts for shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$.

Many existing OOD generalization methods presuppose a particular causal structure among $U, \mathbf{X}, Y$ that guarantees a specific invariance. When $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), the factorization $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y) \mathbb{P}(\mathbf{X} | Y)$ implies that any joint shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is attributable solely to *label shift*, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while the conditional covariate distribution remains invariant, $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ (precisely because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, X8, 79]). Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), the joint shift is driven by *covariate shift*, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while the predictive conditional stays invariant, $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These convenient invariances collapse in the far more common regime in which U causes *both* $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]: here a single shift in $\mathbb{P}(U)$ can trigger label shift, covariate shift, conditional covariate shift, and concept shift *simultaneously*. What survives is a single invariance

Table 1: Types of distribution shift between environments e and e' , with their defining inequalities in the observed distributions of covariates $\mathbf{X}$ and target Y .

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

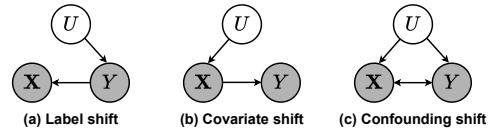


Figure 1: Causal graphs underlying the distribution shifts of Table 1: (a) $U \rightarrow Y \rightarrow \mathbf{X}$ yields label shift $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$ while $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ (precisely because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, X8, 79]). Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), the joint shift is driven by *covariate shift*, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while the predictive conditional stays invariant, $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These convenient invariances collapse in the far more common regime in which U causes *both* $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]: here a single shift in $\mathbb{P}(U)$ can trigger label shift, covariate shift, conditional covariate shift, and concept shift *simultaneously*. What survives is a single invariance

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

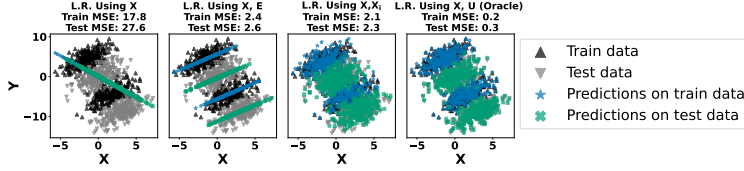


Figure 2: Four linear-regression (L.R.) models under hidden confounding shift with minimal environment overlap (distant μ_e), arranged as a progression of remedies. (i) Trained on X alone, the model recovers a relationship of the wrong sign—Simpson’s paradox. (ii) Augmenting X with environment-specific summary statistics E recovers the correct relationship, though its representational capacity remains limited. (iii) Adding an informative covariate X_i that proxies the confounder U further improves OOD generalization. (iv) The oracle, trained on X and the true U , marks the attainable ceiling.

153 conditional on the confounder, $\mathbb{P}^e(\mathbf{X}, Y \mid U) = \mathbb{P}^{e'}(\mathbf{X}, Y \mid U)$ —the *stable confounding shift*
154 condition [85, 4]—but because U is unobserved, this invariance is inaccessible from the observed
155 data, which is exactly what makes confounding shift so difficult to address.

156 **Information-theoretic quantification of shift.** Following Federici et al. [22], we quantify distri-
157 bution shift with mutual information. Introducing the environment variable E , the quantities
158 $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E \mid Y)$, $I(Y; E \mid \mathbf{X})$, and $I(\mathbf{X}, Y; E)$ measure, respectively, the shifts
159 in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} \mid Y)$, $\mathbb{P}(Y \mid \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$ across environments. In these terms, existing
160 methods encode their structural assumptions as vanishing mutual information: the label-shift assump-
161 tion sets $I(\mathbf{X}; E \mid Y) = 0$, and the covariate-shift assumption sets $I(Y; E \mid \mathbf{X}) = 0$. Under hidden
162 confounding shift, however, *both* of these quantities are generically nonzero—the assumptions on
163 which the corresponding methods rest are violated simultaneously.

164 **The core challenge, concretely.** Figure 2 distills the difficulty that hidden confounding shift poses
165 for OOD generalization into a single controlled experiment. The outcome Y depends on the observed
166 covariate X and a hidden confounder U , and environment-specific shifts in the mean μ_e of U define
167 the distinct environments. A linear regression trained on X alone infers a relationship of the *wrong*
168 *sign* (first subplot): although Y increases with X within every environment, the pooled model
169 recovers the opposite trend—a textbook instance of Simpson’s paradox [73]. Critically, this failure is
170 not peculiar to ERM: dedicated OOD generalization methods, including IRM [6], VREX [38], and
171 Group DRO [69], exhibit the same pathology, because all of them seek an invariant predictor that the
172 confounding shift renders unattainable. Our theory in § 4 shows that the optimal remedy is to learn
173 *environment-specific* relationships, which restores the correct X – Y relationship; the second subplot
174 confirms this, using environment-specific summary statistics of X (mean, standard deviation, and
175 quantiles) to recover the true trend. This is exactly the *backdoor adjustment* logic of causal-effect
176 estimation [58], with environment-specific information serving as a proxy for the confounder. The
177 residual limitation—the representational capacity of statistics derived from X alone—is likewise
178 addressable: as § 4.3 establishes, supplying additional informative covariates that act as proxies for
179 the unobserved confounder yields a further, substantial improvement (third subplot). An oracle model
180 trained on both X and U , which attains the best achievable mean squared error, calibrates the ceiling
181 of this progression.

182 4 Impact of hidden confounding shifts on OOD generalization

183 This section develops a theoretical account of how hidden confounding shift affects the constituent
184 objectives of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the prediction for an input $\mathbf{X}$, where ϕ
185 is a feature extractor (equivalently, a representation map) and f is a classifier acting on $\phi(\mathbf{X})$. We write
186 $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ for entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ for mutual information.
187 The risk of $(f \circ \phi)$ in environment e is $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ for a loss $\ell(\cdot, \cdot)$,
188 and the aim of OOD generalization is a predictor that performs well on both ID and OOD data. The
189 literature instantiates this aim through distinct objectives, each targeting a different notion of stability.
190 Robust-optimization methods [7, 9, 19, 75, 69] minimize the worst-case training risk, $\mathcal{R}^{\text{Rob}}(f \circ \phi) =$
191 $\max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, over the training environments $\mathcal{E}_{tr}$. Variance risk extrapolation (VREX) [38]
192 minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$,
193 penalizing cross-environment risk variance through the hyperparameter β . Invariant risk minimization
194 (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to the constraint that f be simultaneously optimal in
195 every environment. Empirical risk minimization (ERM), by contrast, pools the training environments

and minimizes the aggregate empirical risk $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ is the observed data in environment e [6, 38].

To expose how hidden confounding shift acts on these objectives, we adopt a unifying lens: the maximization of predictive information $I(Y; \hat{Y})$, with $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity measures how much the prediction $\hat{Y}$ reveals about the true label Y and thus serves as a principled, task-agnostic surrogate for predictive quality. Our analysis rests on two properties of the representation $\phi(\mathbf{X})$ —*informativeness* and *invariance*—both of which are recognized as decisive for OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness captures the information $\phi(\mathbf{X})$ carries about Y within each environment. Minimizing it discards predictive signal, whereas maximizing it can be undesirable in settings such as algorithmic fairness, where $\phi(\mathbf{X})$ might exploit sensitive within-environment information and produce unfair predictions. Such bias can be suppressed by minimizing $I(\phi(\mathbf{X}); E)$, but at a cost to predictive performance, as we demonstrate in § 4.2. In the setting of primary interest here—where E carries no sensitive information—maximizing conditional informativeness is beneficial for generalization, a position we develop and substantiate in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

Invariance measures how consistent the representation $\phi(\mathbf{X})$ remains across environments for a given label Y . Low invariance—large residual environment dependence in $\phi(\mathbf{X})$ —invites overfitting to environment-specific artifacts, whereas high invariance strips out those dependencies and enforces invariant learning. Extending the information-theoretic measures of § 3, we use $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to quantify *feature shift* and *concept shift*, respectively; for brevity, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$ according to context.

4.1 A general decomposition of predictive information

We first fix the causal-graph machinery the decomposition relies on. A causal graph $\mathcal{G}$ comprises nodes, one per random variable, and directed edges denoting direct causal influence. A *path* between nodes X_i and X_j is a sequence of distinct nodes joined by edges; a *directed path* from X_i to X_j (with $i < j$) is one whose arrows all point toward X_j , i.e. $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$, in which case X_i is the *parent* of X_{i+1} , X_{i+1} the *child* of X_i , X_i an *ancestor* of X_j , and X_j a *descendant* of X_i . Every path is built from three elementary configurations: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In chains and forks the endpoints are marginally dependent but conditionally independent given the middle node, $X_i \not\perp X_k$ and $X_i \perp X_k | X_j$; in a collider the endpoints are marginally independent but become dependent once the collider or any of its descendants is conditioned on, $X_i \perp X_k$ and $X_i \not\perp X_k | X_j$. A path between X_i and X_k is *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) it contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants lies in $\mathcal{S}$; a path that is not blocked is *open*. When every path from X_i to X_k is blocked by $\mathcal{S}$, we have $X_i \perp X_k | \mathcal{S}$. These rules are precisely what govern the information flow we track through the decomposition below.

Consider now the predictive information $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$ is produced from the learned representation $\phi(\mathbf{X})$. Figure 3 specifies the causal relationships among $\mathbf{X}$, Y , U , $\phi(\mathbf{X})$, $\hat{Y}$, and E , with U the hidden confounder and E the environment variable that records shifts in $\mathbb{P}(U)$ —so that any two environments differ by a shift in $\mathbb{P}(U)$. Because $\hat{Y}$ is a deterministic function of $\phi(\mathbf{X})$, which is in turn a function of $\mathbf{X}$, the prediction contributes nothing beyond these variables for reasoning about Y : $\hat{Y} \perp Y | \mathbf{X}$ and $\hat{Y} \perp Y | \phi(\mathbf{X})$. We

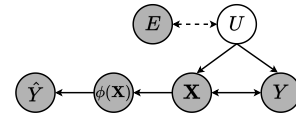


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y encodes that some covariates in $\mathbf{X}$ cause Y while others are caused by Y . The hidden confounder U drives both $\mathbf{X}$ and Y , and the environment E captures shifts in $\mathbb{P}(U)$.

first derive a general decomposition of predictive information that does not specialize the manner in which U influences $\mathbf{X}$ and Y ; the formal treatment of that influence follows in § 4.2.

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs appear in Appendix § A. The decomposition in Equation (1) furnishes a common coordinate system in which existing methods reveal themselves as explicit attempts to maximize or minimize particular terms. IRM [6] constrains a fixed classifier f to be optimal in every environment, which amounts to maximizing $-I(\phi(\mathbf{X}); E | Y)$: any residual environment-specific information in $\phi(\mathbf{X})$ conditional on Y would break the classifier’s simultaneous optimality. DANN [23] and the independence criterion for fair classification [22] instead minimize $I(\phi(\mathbf{X}); E)$, and domain adaptation likewise learns features rendered invariant to the domain [82, 90]. Equation (1) exposes the price of this choice: driving $I(\phi(\mathbf{X}); E)$ down—desirable for fairness—directly lowers $I(Y; \hat{Y})$ and can therefore erode predictive performance [98, 33, 22]. CDAN [51] aligns the joint distributions, $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$, which renders E independent of $(\phi(\mathbf{X}), Y)$ and hence forces $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X}))$, i.e. $I(Y; E | \phi(\mathbf{X})) = 0$; thus, although CDAN never names this conditional mutual information in its loss, its alignment objective drives it to zero. Finally, while $I(Y; E)$ is constant with respect to the learner, a separate family of methods is engineered to be robust to label-distribution shift [69, 95]. The decomposition thus organizes an otherwise heterogeneous literature under a single accounting of predictive information.

Equation (1) is agnostic to the mechanism by which U acts on $\mathbf{X}$ and Y ; specifying that mechanism sharpens which terms it is principled to control. If $U \rightarrow Y$, $U \not\rightarrow \mathbf{X}$, and $Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, so minimizing $I(\phi(\mathbf{X}); E | Y)$ is well motivated. Symmetrically, if $U \not\rightarrow Y$, $U \rightarrow \mathbf{X}$, and $\mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, motivating the minimization of $I(Y; E | \phi(\mathbf{X}))$. The genuinely challenging—and, as we argued, the most prevalent—regime is $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$: here the terms in Equation (1) interact non-trivially and no single term can be targeted in isolation. The next section resolves exactly this case.

4.2 Predictive information under hidden confounding shift

To determine how a hidden confounder U acting on both $\mathbf{X}$ and Y shapes the predictive information, we analyze the *two exhaustive orientations* of the $\mathbf{X}$ – Y relationship: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. Under $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow directly from the d-separation rules: conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ at the collider Y , admitting extra information flow from $\phi(\mathbf{X})$ to E via Y , whereas conditioning on $\phi(\mathbf{X})$ partially blocks the flow from Y to E at $\mathbf{X}$ to the extent that $\phi(\mathbf{X})$ retains information about $\mathbf{X}$.

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \quad \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Under $Y \rightarrow \mathbf{X}$, the inequalities in (3) follow by the same reasoning with the roles reversed: conditioning on Y blocks the flow from $\phi(\mathbf{X})$ to E at Y , while conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ at $\mathbf{X}$ —because $\phi(\mathbf{X})$ is a child of the collider $\mathbf{X}$, and conditioning on the child of a collider opens the path through it [58] (cf. the preliminaries in § 4.1).

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \quad \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Combining the inequalities in (2) and (3) collapses the general decomposition into a strikingly simple form.

Proposition 4.2. For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if

294 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

295 Proposition 4.2 treats the two orientations $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, but the former dominates
 296 real-world tabular prediction: 11 of the 16 datasets in recent benchmark studies [57, 50, 24] conform
 297 to the structure $\mathbf{X} \rightarrow Y$, and in the remaining 5 the covariates caused by Y are far fewer than those
 298 causing Y . The result therefore covers the overwhelming majority of settings of practical interest. Its
 299 central consequence is the following.

Under hidden confounding shift—that is, in the special cases (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ —the four terms *variation, label shift, feature shift, and concept shift* that appear in (1) cancel exactly, and (4) reduces to a single objective: the predictive information is maximized precisely by maximizing the difference *conditional informativeness* – *residual*. Invariance-based control of the shift terms is thus not merely secondary but immaterial in this regime.

300

301 **Maximizing conditional informativeness.** A recent mixture-of-experts (MoE) model for OOD
 302 generalization under hidden confounding shift [62] assigns one expert to each value of the hidden
 303 confounder, so that every expert maximizes performance within the environment induced by that
 304 value. This construction is exactly a mechanism for maximizing the conditional informativeness
 305 $I(\phi(\mathbf{X}); Y | E)$, and Equation (4) therefore supplies its theoretical justification. The same equation,
 306 however, also delineates the limits of such models and the need for methods that hold under more
 307 general confounding shift: the approach of Prashant et al. [62] requires overlapping confounder
 308 support, $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, together with discrete-valued proxies.

Figure 2 shows that even without confounder-support overlap and with a continuous-valued proxy—both assumptions relaxed relative to prior work—a linear regression furnished with environment-specific information recovers the correct covariate–target relationship across environments. This concrete demonstration motivates OOD generalization methods that remain effective precisely where confounder support is limited and the confounder is continuous-valued.

309

310 **Minimizing the residual.** The residual $I(\phi(\mathbf{X}); Y | \hat{Y})$ measures the information $\phi(\mathbf{X})$ retains
 311 about Y beyond what the prediction $\hat{Y}$ already conveys, and admits the identity $I(\phi(\mathbf{X}); Y | \hat{Y}) =$
 312 $H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ is in turn tied to the expected
 313 cross-entropy loss through $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29], where
 314 $H(Y | \hat{Y})$ —the *refinement error*—quantifies the model’s ability to discriminate between classes.
 315 Minimizing the residual therefore admits two complementary routes. The first drives $I(\phi(\mathbf{X}); Y | \hat{Y})$
 316 to zero directly by rendering $\phi(\mathbf{X})$ conditionally independent of Y given $\hat{Y}$, i.e. $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. The
 317 second minimizes the refinement error $H(Y | \hat{Y})$ itself—for example, via training schemes that first
 318 reduce refinement error for accurate prediction and only then address calibration error [10].

319 4.3 Impact of additional informative variables

320 Whereas § 4.2 characterizes the optimal generalization strategy when the confounder is entirely
 321 unobserved, practice often affords *proxies* for the hidden confounder [4, 85, 62] that can stand in
 322 for it. We now quantify precisely how adding such non-causal but informative covariates moves the
 323 predictive information.

324 **Definition 4.3** (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,*
 325 *neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are*
 326 *not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

327 Because $U \rightarrow Y$, any covariate informative of U is equally informative of Y , and conversely. The
 328 decomposition (1) shows that maximizing $I(Y; \hat{Y})$ demands, among other things, minimizing the

Table 2: Distribution shift, quantified by mutual information with the environment E (mean $\pm$ standard deviation over 10 random subsets of 40,000 samples). Across all eight datasets, conditional covariate shift $I(\mathbf{X}; E | Y)$, label shift $I(Y; E)$, covariate shift $I(\mathbf{X}; E)$, and concept shift $I(Y; E | \mathbf{X})$ are all present and, by one-sample t -tests against zero ($\alpha = 0.05$), all significantly nonzero (p-value ≈ 0)—the joint signature predicted for hidden confounding shift.

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

concept shift $I(Y; E | \phi(\mathbf{X}))$; consistent with this, Liu et al. [50] report empirically that collecting additional covariates $\mathbf{X}_I$ for which $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ is more stable across environments—that is, reducing concept shift—improves OOD test accuracy. We now place this observation on a firm theoretical footing, proving that informative variables raise predictive information by moving the individual terms of (1) in their beneficial directions.

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned}
 (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\
 (iii) \quad & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)
 \end{aligned} \tag{5}$$

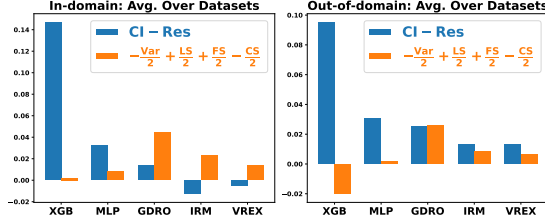
Proposition 4.3 establishes that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii) while decreasing concept shift (5.iii)—each a movement in the direction that raises predictive information. The one countervailing effect is that variation also grows (5.iv), so realizing the full benefit of informative covariates may call for dedicated strategies to control this variation term.

5 Experimental results

We evaluate our theory on both real-world and synthetic data, addressing three questions in turn: (i) whether hidden confounding shift is actually present in real-world data, (ii) how the individual terms of the decomposition (1) govern OOD generalization under such shift, and (iii) whether informative covariates improve generalization as the theory predicts. The real-world evaluation spans eight tabular benchmarks—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—with datasets and domain splits taken from the *TableShift* benchmark [24], and we report both ID and OOD test accuracy. The method suite is deliberately broad, covering all three families identified in § 4: two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. Each reported value is the mean over 5 random hyperparameter draws (Appendix B); random seeds have a statistically insignificant effect on these datasets [24]. Full setup details appear in Appendix § B, and results under explicit hyperparameter tuning in Appendix C.

Hidden confounding shift is pervasive. As established in § 3, hidden confounding shift leaves a characteristic signature: it induces observable shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$ simultaneously. Table 2 confirms this signature directly: label shift, covariate shift, conditional covariate shift, and concept shift are all present, and all statistically significant, in every real-world dataset—exactly the fingerprint of a shifting hidden confounder rather than any single isolated shift type. A complementary qualitative analysis, in which we query GPT-4o [1] for plausible hidden confounders in each benchmark [24, 57] (Appendix § E), corroborates this reading; in *food stamps*, for example, state-specific *economic policies* plausibly drive both *household income* and *food stamp reciprocity*.

Conditional informativeness tracks accuracy. Proposition 4.2 predicts that, under hidden confounding shift, the difference *conditional informativeness* – *residual* is the sole driver of predictive infor-



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Across the eight datasets, the difference *conditional informativeness* – *residual* (plots) is positively correlated with average ID and OOD test accuracy (table, right). XGB attains both the largest difference and the highest accuracy (85.22% ID, 77.00% OOD), while the invariance-oriented methods IRM and VREX, which suppress the shift terms rather than maximize conditional informativeness, trail the ERM-based baselines.

Table 3: Sign consistency of each decomposition term with accuracy gains as the covariate set grows ($C \subseteq AC \subseteq A$), together with ID- and OOD-test accuracy per covariate set. Concept shift (**CS**) shows high sign consistency for every method, but conditional informativeness (**CI**)—the term Proposition 4.2 identifies as governing generalization—is where the methods diverge, and XGB attains the highest CI consistency (0.89) and the best accuracy in every covariate setting. **Res**: residual; **Var**: variation; **FS**: feature shift; **C**: causal, **AC**: arguably causal, **A**: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

mation. Figure 4 tests this prediction on the six datasets of Table 2 that possess more than one training environment [57, 24] (dataset-wise results in Appendix C), and the prediction holds cleanly: the difference is positively correlated with both ID and OOD accuracy, and XGB—the method that attains the largest value of *conditional informativeness* – *residual*—is also the strongest performer, at 85.22% ID and 77.00% OOD accuracy. The theory’s more distinctive prediction is likewise borne out empirically: in the dataset-wise results (Appendix C), the orange bars collapse toward zero, because under hidden confounding shift the aggregate $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes exactly as Proposition 4.2 dictates. The resulting method ranking reproduces the relative ID/OOD ordering reported in prior work [57, 24].

Informative covariates improve generalization. We next examine how enlarging the covariate set improves OOD generalization, using the nested partition of [57], $\text{causal} \subseteq \text{arguably causal} \subseteq \text{all covariates}$ (Appendix B), and tracking how the terms of (1) respond as the covariate set grows. To make this response legible, we introduce the *sign consistency metric*: on each transition to a larger covariate set, a positively weighted term (e.g. conditional informativeness) is counted as consistent when its value rises together with accuracy, and a negatively weighted term (e.g. concept shift) when its value falls as accuracy rises. The metric is the fraction of transitions in which a term moves in its *beneficial* direction, and thus reports how reliably that term contributes to better generalization (formal definition in Appendix B). Table 3 shows that both conditional informativeness and concept shift attain high sign consistency with accuracy gains. The decisive distinction is that, although every method benefits from reduced concept shift, only XGB additionally amplifies conditional informativeness—precisely the term Proposition 4.2 identifies as governing predictive information—which accounts for its consistent accuracy advantage. Dataset-specific results are provided in Appendix C.

Because real-world data offer no guarantee that *all available covariates* in fact contain *every relevant or sufficiently informative* variable, we corroborate the hypothesis on synthetic data whose causal structure is known by construction: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, with $\mathbf{X}_I$ the informative covariates generated by the hidden confounder U . Introducing $\mathbf{X}_I$ produces exactly the pattern Proposition 4.3 predicts—increased conditional informativeness and feature shift alongside reduced concept shift (Figure 5)—and the variation term stays identically zero, as expected given the simplicity of the generative process. The data-generating process and further results appear in Appendix D.

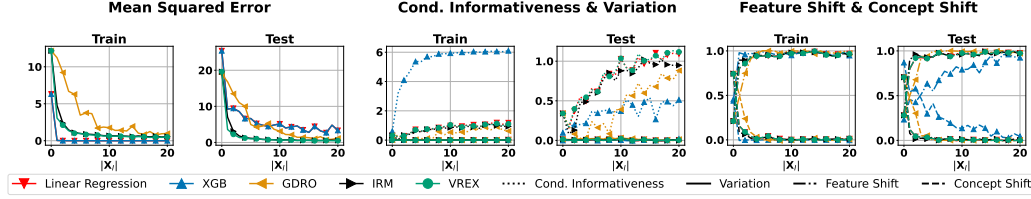


Figure 5: On synthetic data, adding proxy variables $\mathbf{X}_I$ of U that are informative of Y reduces MSE while increasing conditional informativeness and feature shift and reducing concept shift—the exact term movements predicted by Proposition 4.3. The variation term remains zero throughout.

6 Discussion, limitations, and future work

This work resolves a standing puzzle in OOD generalization by decomposing the predictive information between model outputs and true outcomes into interpretable terms (Equation (1)) and reading off, from that decomposition, exactly what limits generalization. Our central finding is that hidden confounding shift—registered as coupled changes in the covariate and outcome distributions across environments—is unavoidable, and that no finite set of observed covariates or proxies can fully dissolve the ambiguity it induces. Because confounding entangles covariate, label, concept, and feature shift at once (inequalities (2) and (3)), any strategy that targets these shifts in isolation is inherently inadequate; we prove instead that learning environment-specific input–output mappings strictly dominates the enforcement of a single invariant predictor (Equation (4)), and our experiments on eight real-world benchmarks and controlled synthetic data confirm this account. In establishing both the inevitability of hidden confounding and the necessity of confronting it head-on, this work supplies the theoretical foundation and the empirical evidence for a next generation of robust OOD methods. The agenda it opens is concrete: (i) algorithms that remain effective under limited confounder overlap, (ii) a quantitative characterization of the cost–accuracy trade-off in acquiring informative covariates, (iii) principled handling of entangled shifts without recourse to untestable proxy assumptions, and (iv) explicit modeling of the ambiguity introduced by unobserved confounders as the basis for new OOD-robust paradigms.

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693 Appendix

694 A Proofs of propositions

695 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 696 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 697 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

698 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 699 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

700 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 701 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 702 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 703 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

704 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

705 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

706 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 707 expansion instead of $\phi(\mathbf{X})$. $\square$

708 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 709 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 710 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

711 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 712 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

713 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 714 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

715 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 716 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 717 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

718 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

719 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

720 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

721 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 722 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 723 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

724 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 725 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

726 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

727 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

728 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

729 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 730 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

731 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 732 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

733 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
734 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
735 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
736 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

737 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
738 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
739 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
740 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
741 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

742 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

743 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

744 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
745 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
746 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
747 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
748 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

749 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
750 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
751 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
752 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
753 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

754 B Experimental setup

755 This section specifies the experimental setup in full. Because random seeds exert a statistically
756 insignificant effect in most of these settings [24], the main-paper results report the mean and stan-
757 dard deviation across five hyperparameter values sampled from the ranges in Table B1 for each
758 dataset–method combination. The mutual-information terms in the decomposition are estimated with
759 the nonparametric entropy-estimation toolbox [37, 77, 78], and, following [24], are evaluated on both
760 in-distribution (ID) test data drawn from the training domains and out-of-distribution (OOD) test data
761 drawn from the held-out test domains.

762 For consistency, we draw 20,000 random sam-
763 ples from each of the ID and OOD test sets, ex-
764 cept where a dataset contains fewer than 20,000
765 samples. Sample size exerts a negligible influ-
766 ence on the estimated mutual-information terms
767 throughout: as Figure B1 illustrates, the terms of
768 Equation (1) computed for the XGB representa-
769 tions on the *income* dataset remain stable as the
770 sample size varies. The representation $\phi(\mathbf{X})$ is
771 taken from the layer immediately preceding the
772 classification head; for XGBoost, it is formed by
773 concatenating the model’s output margins with
774 its SHAP values. All experiments ran on a single
775 NVIDIA RTX A6000 GPU.

775 For the experiments on the impact of informative covariates, we adopt the covariate taxonomy
776 of Nastl and Hardt [57], which partitions the covariates into *causal*, *arguably causal*, and *all covari-*
777 *ates*. Causal covariates exert an established influence on the target, with well-supported directionality
778 and negligible risk of reverse causation; restricting to them, however, risks omitting genuine parents
779 that remain unknown. Arguably causal covariates carry uncertain causal status, satisfying at least one
780 of the following: (1) being known to be causal, (2) exerting a plausible but possibly bidirectional
781 influence, or (3) having a likely though unconfirmed causal effect. Both groups approximate the true
782 causal parents to the extent that current knowledge allows, and their relationships may themselves be

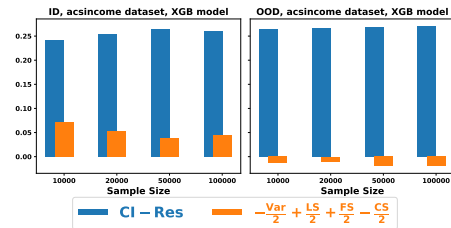


Figure B1: The evaluated predictive-information terms remain stable across sample sizes on the *income* dataset, confirming that the reported estimates are not artifacts of a particular subsample size.

Table B1: Hyperparameters and their search ranges for each method evaluated in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

783 confounded. Some datasets further contain *anti-causal* covariates, which the target likely influences
784 but not conversely, while the *all covariates* set comprises every observed variable irrespective of
785 causal status. The sign consistency metric of § 5 is defined as follows. For each $(m, \sigma_m) \in M$, where
786 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
787 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, and for a fixed dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

788 Here P is the set of covariate-set pairs (s, t) with $t \supset s$ —for instance, s the causal and t the arguably
789 causal covariates—and the factor of 2 in the denominator $2|P|$ accounts for the two evaluation

splits, train and test. The term $\mu_s^{\text{tr/te}}(m)$ denotes the mean of measure m for covariate set s on the training/testing split, $\mathbb{I}(\cdot)$ is the indicator function, and $|P| = 3$ for the covariate settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$. The final sign consistency value for each m is obtained by averaging across datasets, $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, with $|D|$ the number of datasets.

C Additional results on real-world datasets

This section presents additional real-world results that reinforce the main-paper findings at the level of individual datasets. Figures C2 and C3 compare, per dataset, the two composite quantities *conditional informativeness – residual* and $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ (their aggregate appears in main-paper Figure 4, left). Two consistent patterns emerge. First, in line with Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall model performance. Second, and more tellingly, the difference *conditional informativeness – residual* contributes substantially more to the predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ —the signature of hidden confounding shift, under which the latter aggregate vanishes. This directly motivates maximizing conditional informativeness as the operative objective for OOD generalization under hidden confounding shift.

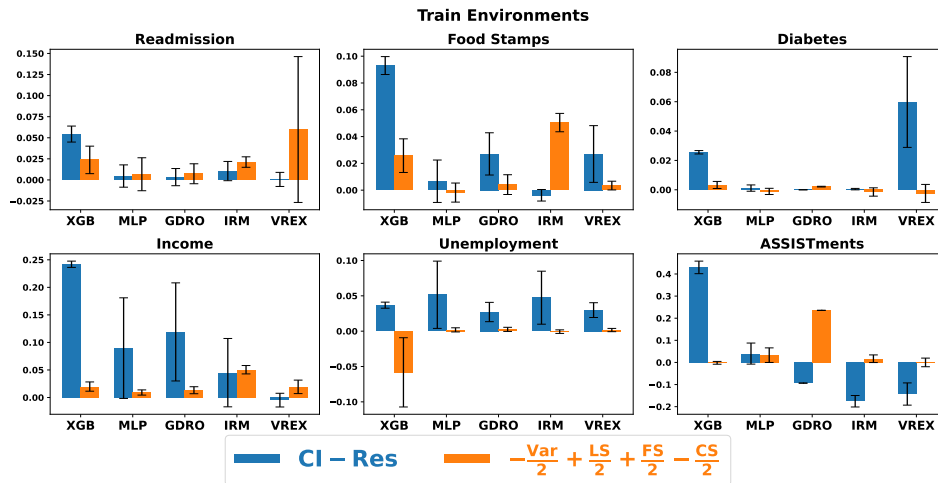


Figure C2: Dataset-specific decomposition on ID data, disaggregating the aggregate of Figure 4. The difference *conditional informativeness – residual* dominates the predictive information across datasets.

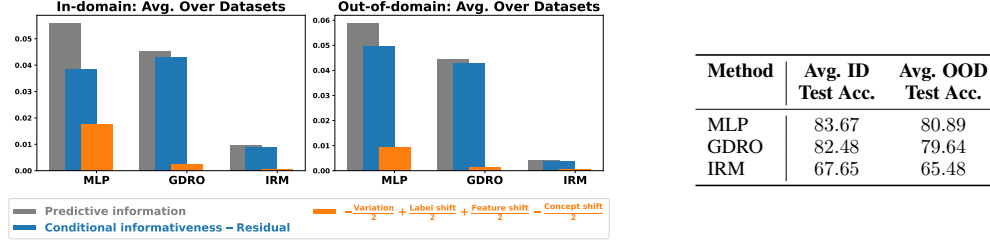


Figure C4: Results under explicit hyperparameter tuning (best ID-validation configuration). The difference *conditional informativeness* – *residual* (plots) remains positively correlated with average ID and OOD test accuracy over the five datasets (table, right), and continues to dominate $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ in its contribution to predictive information—confirming that the main-paper conclusions are robust to hyperparameter selection.

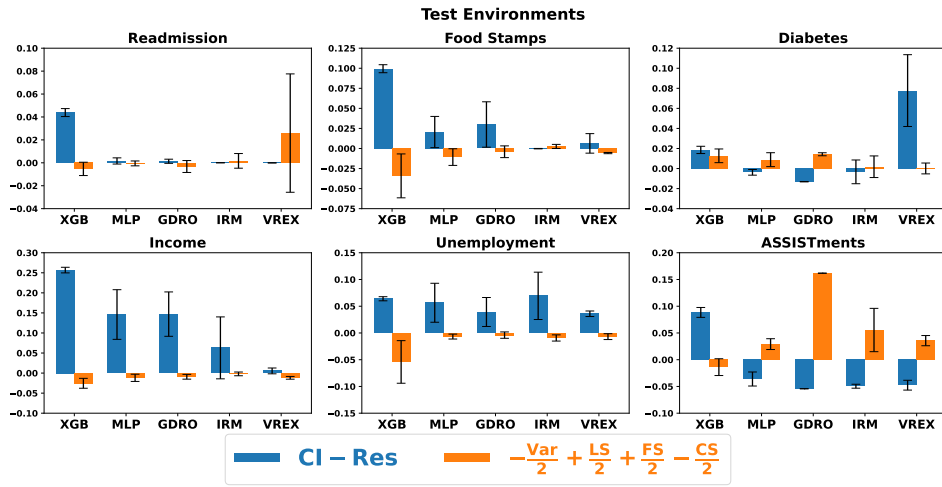


Figure C3: Dataset-specific decomposition on OOD data. As on ID data, the difference *conditional informativeness* – *residual* is the dominant contributor to predictive information, consistent with hidden confounding shift.

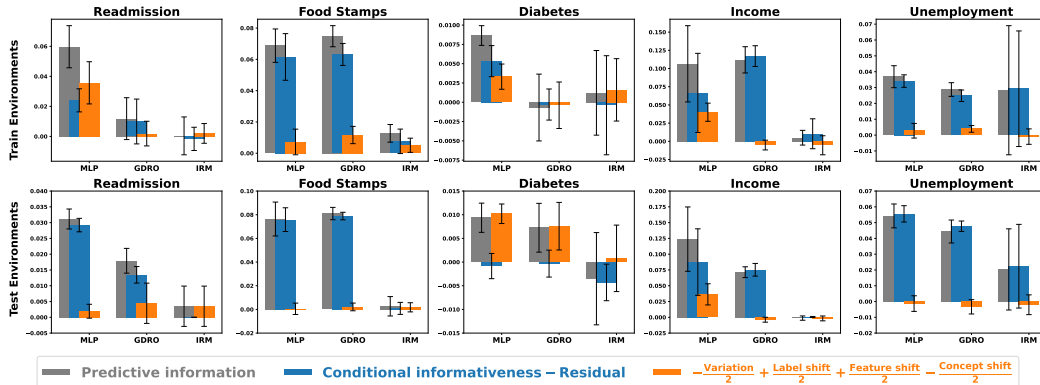


Figure C5: Dataset-specific comparison of the predictive-information terms under hyperparameter tuning, mirroring the aggregate trends of Figure C4.

Table C2: Dataset-specific ID and OOD test accuracies (mean $\pm$ standard deviation) under hyperparameter tuning, corroborating the aggregate ranking in Figure C4.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 $\pm$ 0.14	61.75 $\pm$ 0.22	84.74 $\pm$ 0.03	82.00 $\pm$ 0.06	87.66 $\pm$ 0.02	83.22 $\pm$ 0.05
GDRO	60.47 $\pm$ 1.08	57.60 $\pm$ 0.42	84.47 $\pm$ 0.04	81.41 $\pm$ 0.09	87.47 $\pm$ 0.10	82.90 $\pm$ 0.10
IRM	50.28 $\pm$ 7.35	50.85 $\pm$ 1.74	80.91 $\pm$ 0.00	78.01 $\pm$ 0.00	42.54 $\pm$ 36.5	43.48 $\pm$ 31.9

	Income		Unemployment	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	82.92 $\pm$ 0.05	81.46 $\pm$ 0.01	97.27 $\pm$ 0.02	96.04 $\pm$ 0.05
GDRO	82.70 $\pm$ 0.05	80.16 $\pm$ 0.45	97.28 $\pm$ 0.01	96.10 $\pm$ 0.00
IRM	67.91 $\pm$ 0.00	60.20 $\pm$ 0.00	96.61 $\pm$ 0.00	94.84 $\pm$ 0.00

Figures C6–C15 report the full per-method, per-dataset breakdown of the decomposition terms in (1), and they make the mechanism behind the main-paper results directly visible. The **variation** (second column) and **feature shift** (third column) terms are comparable in magnitude, so by Equation (1) their difference is near zero and contributes negligibly; the same holds for **label shift** (fourth column) and **concept shift** (fifth column). Consequently, in nearly every setting the difference *conditional informativeness* - *residual* is the majority contributor to predictive information—precisely the collapse to Equation (4) that Proposition 4.2 predicts under hidden confounding shift.

Hyperparameter tuning. To confirm that our conclusions do not depend on the random-hyperparameter protocol used in the main paper, we additionally perform explicit tuning: a grid search that selects the best configuration by ID-validation accuracy (never OOD-validation accuracy), comprising 1,500 experiments (5 datasets $\times$ 3 models $\times$ 100 sampled hyperparameter configurations). Each experiment is repeated over 5 random seeds to obtain means and standard deviations for both accuracies and mutual-information terms, despite the seed effect being insignificant in most cases. The tuned results (Figure C4) reproduce the main-paper trends exactly: by Proposition 4.1 the sum of the **blue** and **orange** terms correlates positively with overall performance, and the difference *conditional informativeness* - *residual* again contributes far more to predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. The dataset-specific results in Figure C5 and Table C2 exhibit the same behavior.

Scope of the predictive-information analysis. We state plainly the boundary of the decomposition-based analysis. Predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, not accuracy: a binary classifier that systematically inverts the label (predicting $\hat{Y} = 1$ when $Y = 0$) attains the maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation yet achieves 0% accuracy, because mutual information quantifies *reduction in uncertainty* rather than correctness. The individual terms carry analogous caveats—conditional informativeness $I(\phi(X); Y | E)$ can mask a harmful variation term $I(\phi(X); E | Y)$, and feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ via spurious correlation. These caveats do not undermine the analysis so much as delimit its intended use: for the reasonably performant models our experiments consider, the decomposition reliably distinguishes predictive information arising from generalizable structure from that arising from environment-specific artifacts, and it does so while remaining interpretable—yielding actionable, targeted directions for improvement.

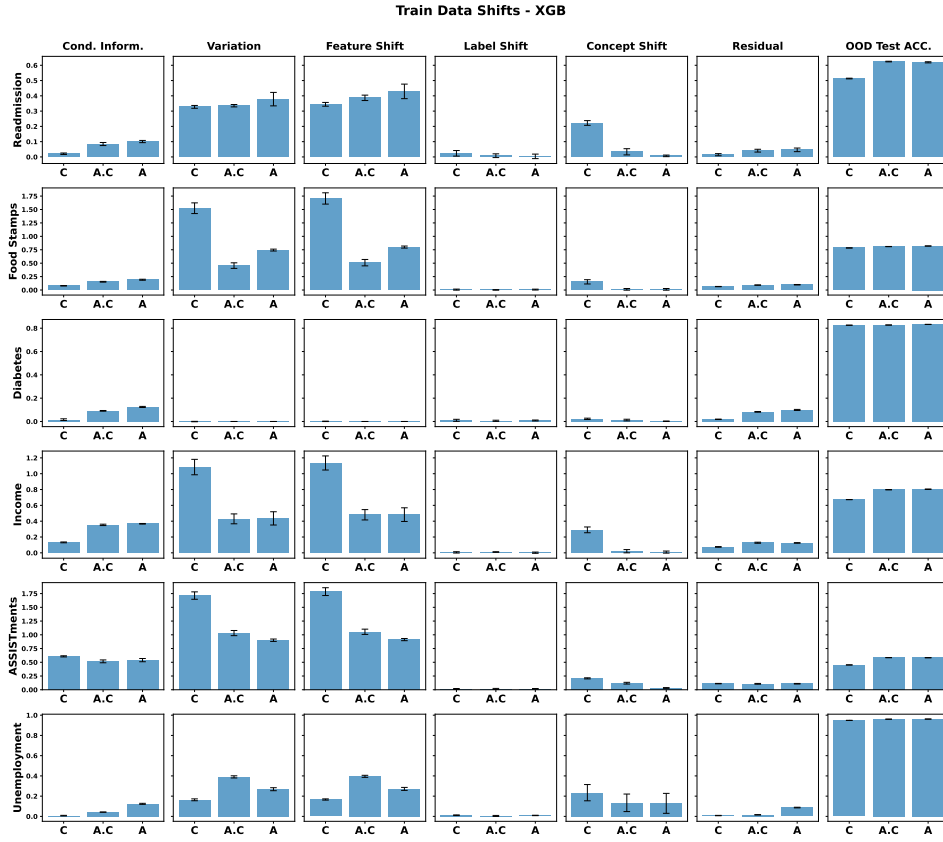


Figure C6: Term-wise decomposition of predictive information on ID (train-domain) data for XG-Boost across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

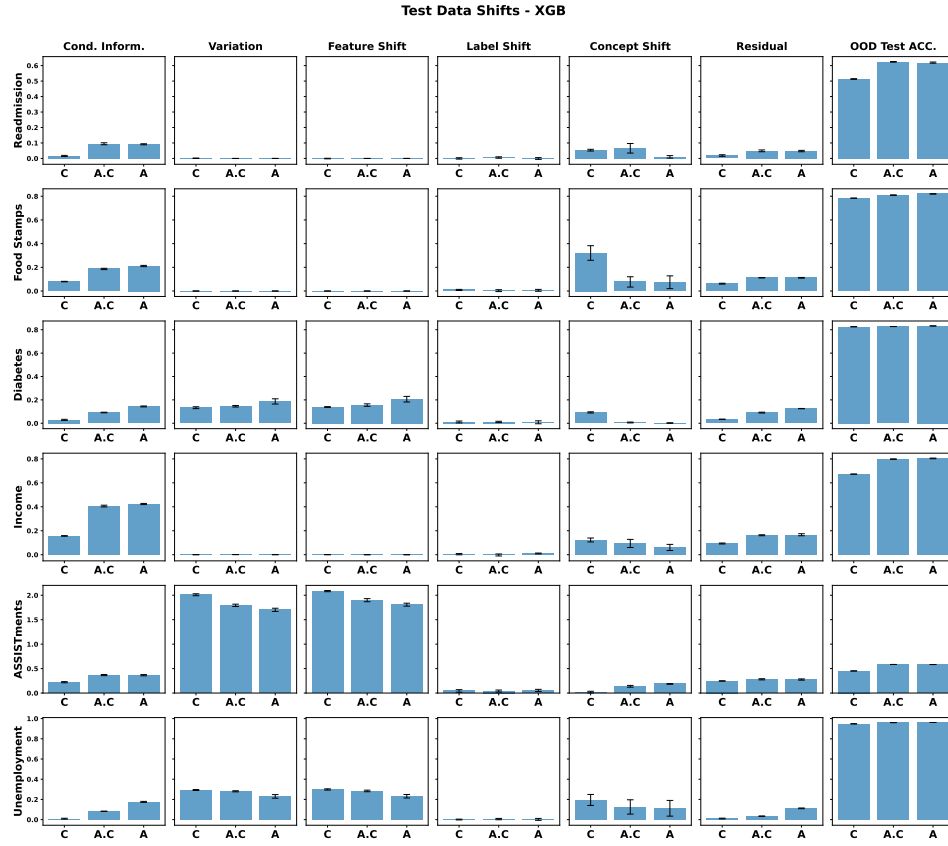


Figure C7: Term-wise decomposition of predictive information on OOD (test-domain) data for XGBoost across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

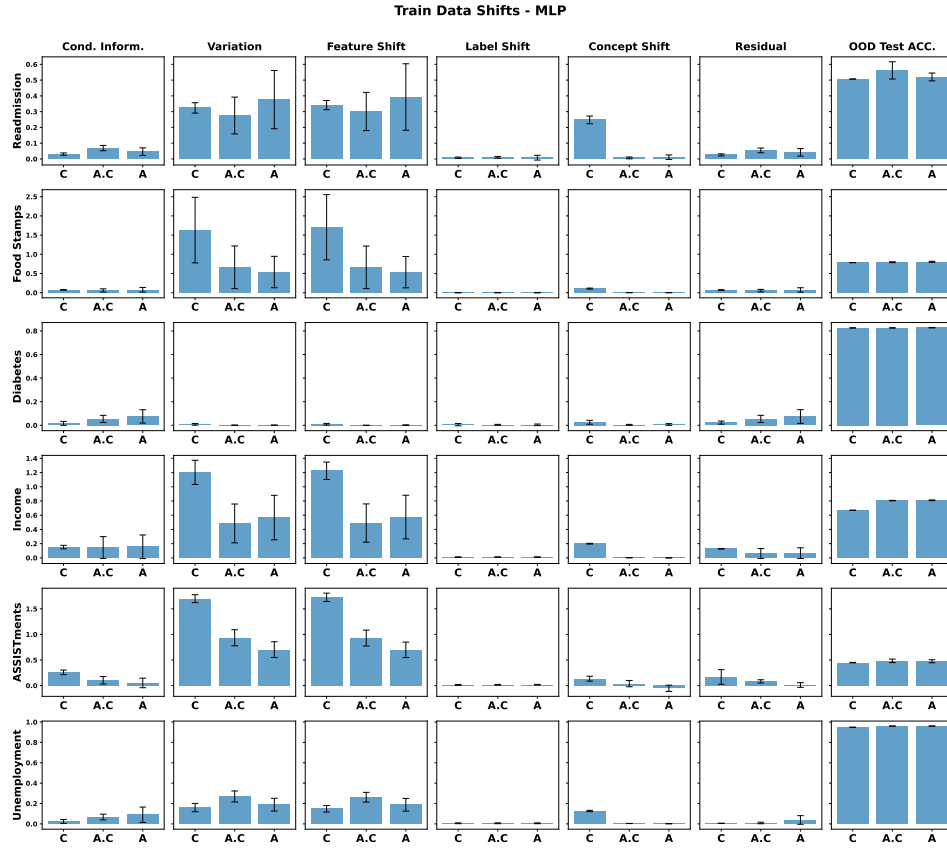


Figure C8: Term-wise decomposition of predictive information on ID (train-domain) data for MLP across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

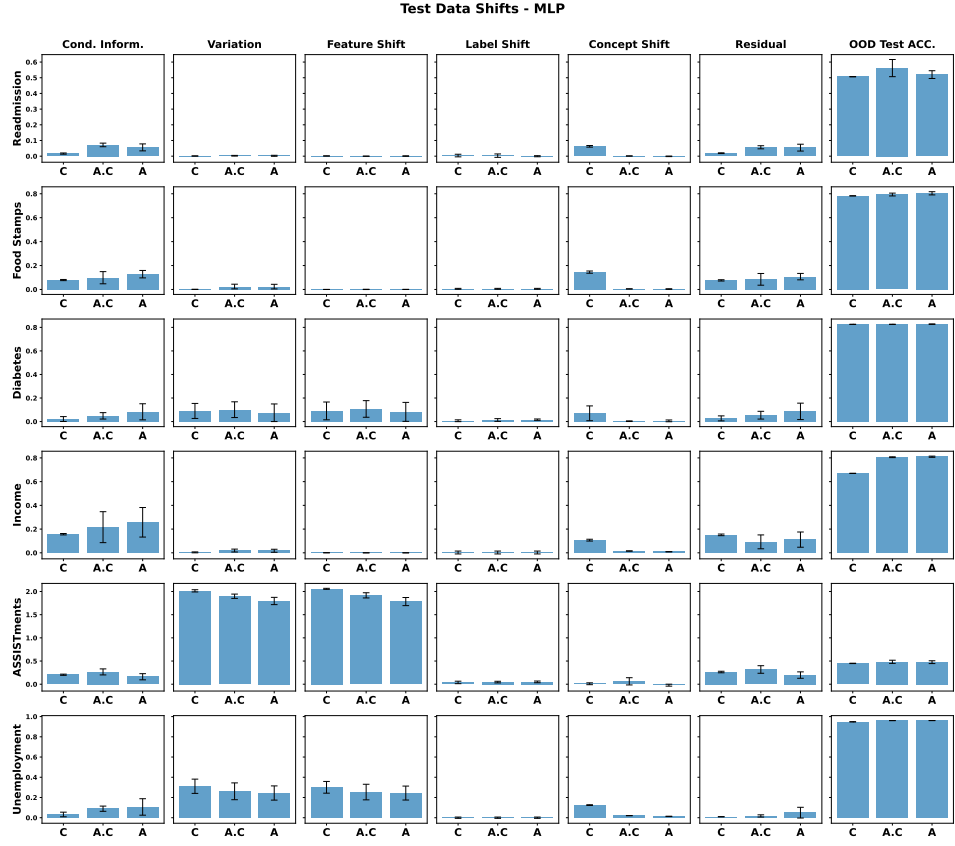


Figure C9: Term-wise decomposition of predictive information on OOD (test-domain) data for MLP across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

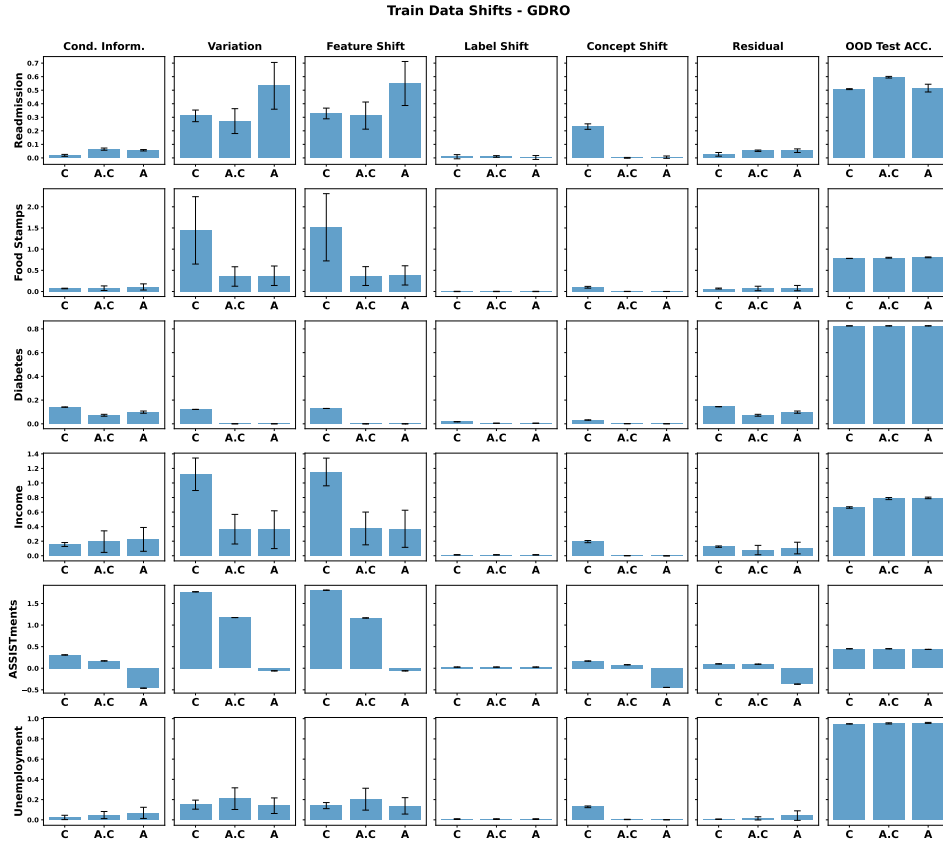


Figure C10: Term-wise decomposition of predictive information on ID (train-domain) data for GDRO across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

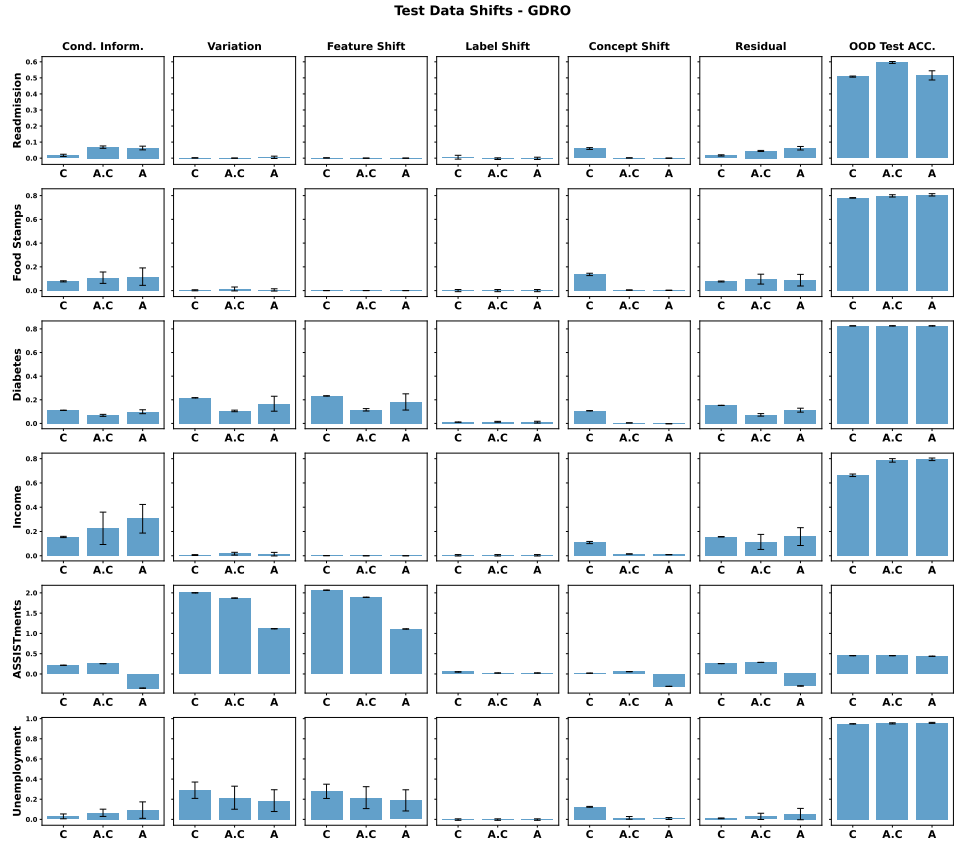


Figure C11: Term-wise decomposition of predictive information on OOD (test-domain) data for GDRO across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

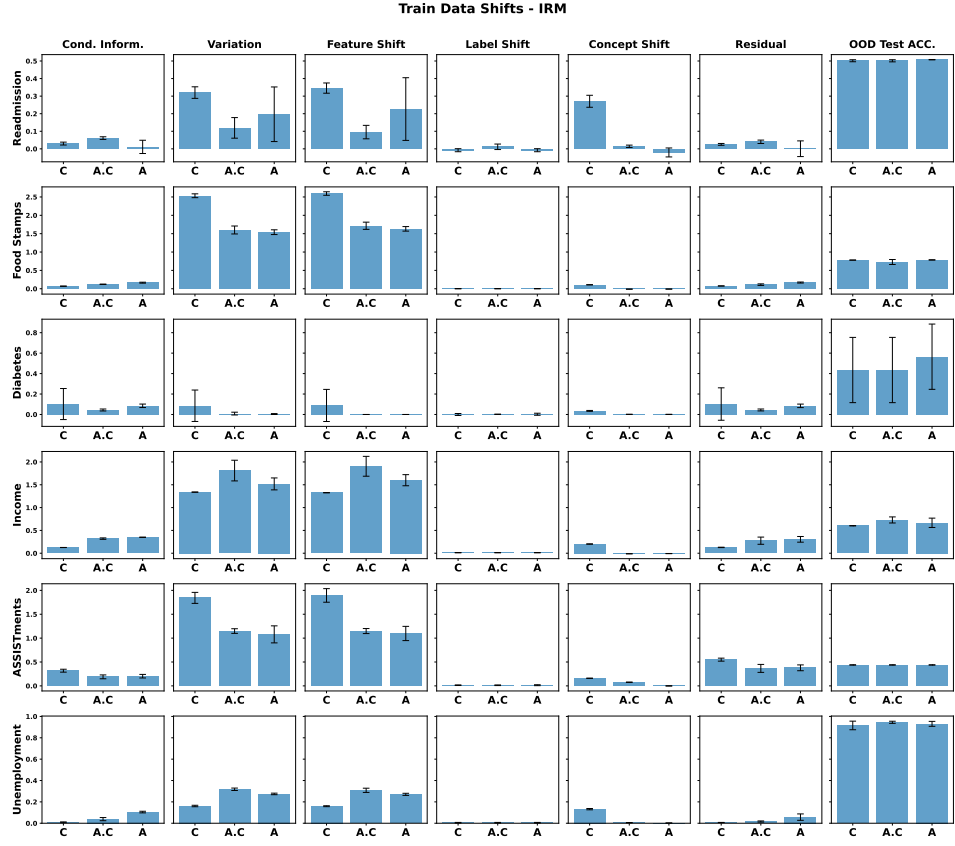


Figure C12: Term-wise decomposition of predictive information on ID (train-domain) data for IRM across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

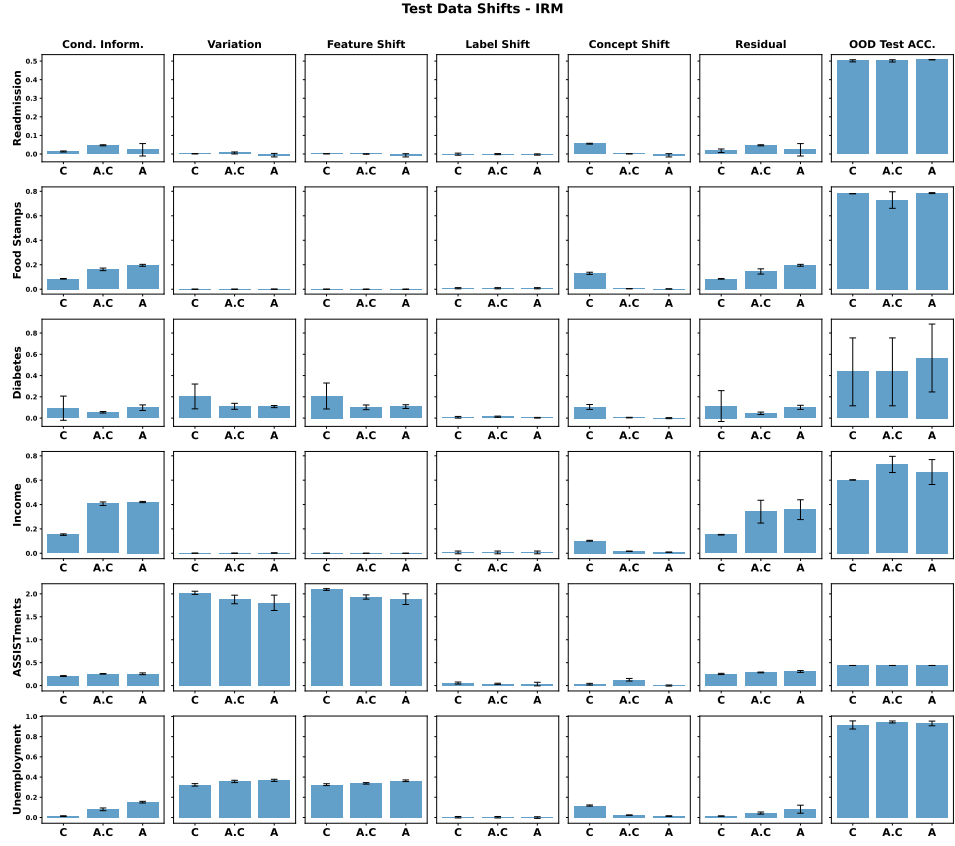


Figure C13: Term-wise decomposition of predictive information on OOD (test-domain) data for IRM across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

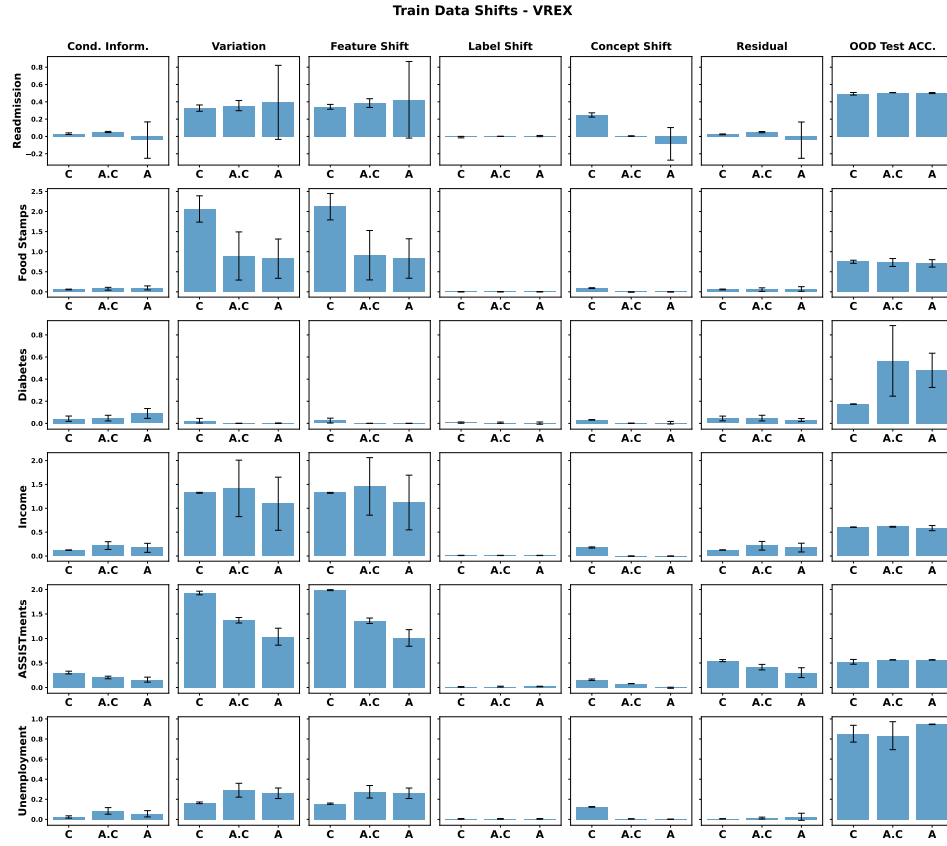


Figure C14: Term-wise decomposition of predictive information on ID (train-domain) data for VREX across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

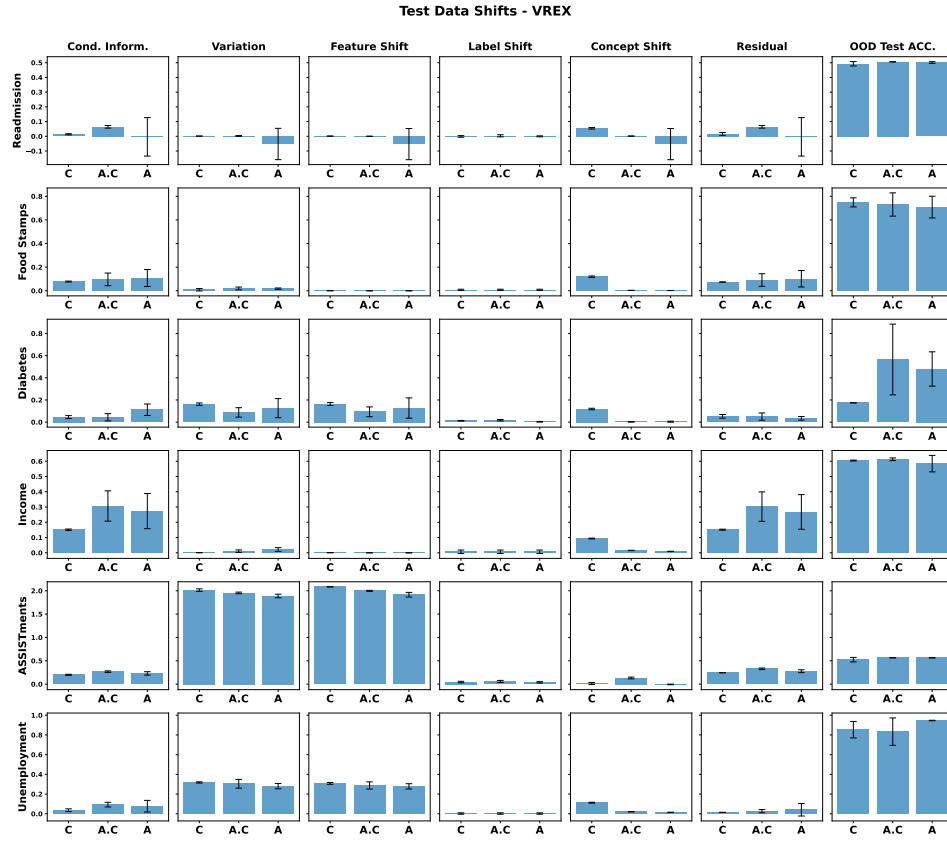


Figure C15: Term-wise decomposition of predictive information on OOD (test-domain) data for VREX across the nested feature groups: (C) causal, (A.C) arguably causal, (A) all.

837 D Results on synthetic datasets

838 The synthetic experiments isolate three questions that real-world data cannot cleanly resolve: (i) the
 839 effect of different informativeness criteria, (ii) the dependence of performance on the degree of
 840 confounder overlap across environments, and (iii) the influence of the ratio of causal to anti-causal
 841 variables. Working with synthetic data lets us fix the ground-truth causal structure and vary one factor
 842 at a time. The data-generating process follows the structural equations in (12), with causal structure
 843 $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is the unobserved confounder, X the observed
 844 covariate, Y the target, and $\mathbf{X}_I$ the additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

845 We consider several functional forms for f_X , f_i , and f_Y , reporting results for each. Domain shift is
 846 induced by systematically varying the environment-specific mean μ_u^e of the confounder distribution,
 847 which affords direct control over the degree of distributional shift and hence a controlled test of its
 848 effect on performance across environments.

849 **Informativeness improves accuracy across criteria.** We begin with the linear structural equations
 850 $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$, with training environments $\mu_u^e \in \{-2, 2\}$ and test
 851 environments $\mu_u^e \in \{0, 4\}$. For Figure 5 of the main paper we use one hidden confounder and
 852 $|\mathbf{X}_I| = 20$ informative covariates: increasing the number of informative covariates from 0 to 20
 853 produces exactly the predicted co-movement—(i) reduced mean squared error, (ii) higher conditional
 854 informativeness, (iii) larger feature shift, and (iv) lower concept shift. Two further informativeness
 855 criteria reproduce the same pattern. First, with one hidden confounder and a single informative
 856 covariate ($|\mathbf{X}_I| = 1$), sweeping the noise level $\sigma_i \in \{0, 0.1, \dots, 2.0\}$ shows (Figure D16) that as
 857 noise falls from 2.0 to 0.0 the MSE drops while conditional informativeness and feature shift rise and
 858 concept shift falls, with the variation term identically zero throughout, as in every other synthetic
 859 experiment. Second, with 20 hidden confounders and 20 corresponding informative covariates under
 860 the modified equations $f_X = 0.3U$, $f_i = 0.2U$, $f_Y = X - 2U$, introducing the informative covariates
 861 again yields the same four movements (Figure D17). Across all three criteria, then, informativeness
 862 translates directly into improved OOD generalization, precisely as the theory predicts.

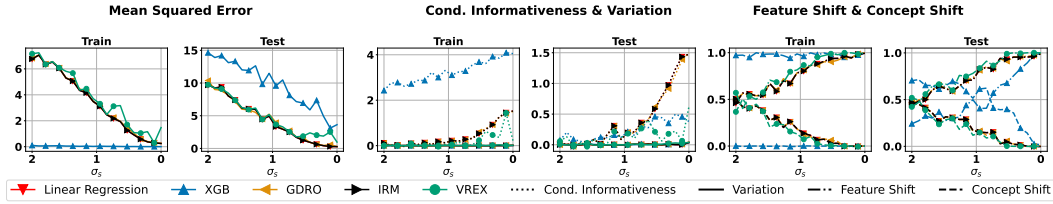


Figure D16: Single informative covariate ($|\mathbf{X}_I| = 1$), varying its noise level σ_i . A lower-noise proxy reduces MSE and increases conditional informativeness and feature shift while reducing concept shift, consistent with Proposition 4.3.

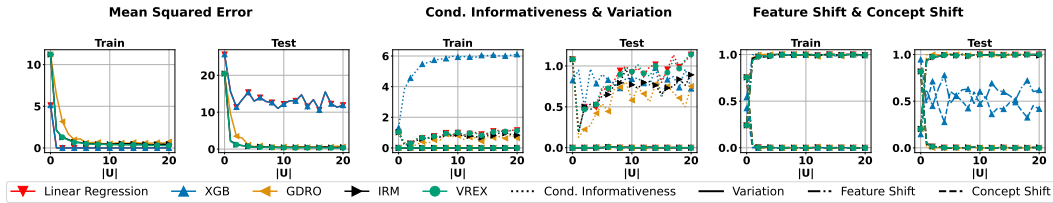


Figure D17: Setting with 20 hidden confounders U and their corresponding informative covariates $\mathbf{X}_I$. Adding proxies informative of Y reduces MSE and increases conditional informativeness and feature shift while reducing concept shift, extending the single-confounder result to the multi-confounder regime.

863 **Robustness to limited confounder overlap.** The method of Prashant et al. [62] assumes that test-
 864 time confounder support lies within training support. Having shown that correct relationships are

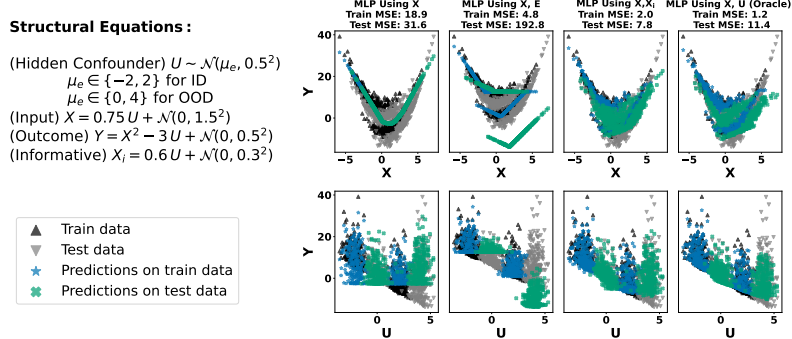


Figure D18: MLP under the low-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). Informative covariates (X_i) yield the largest gains even when confounder support does not overlap.

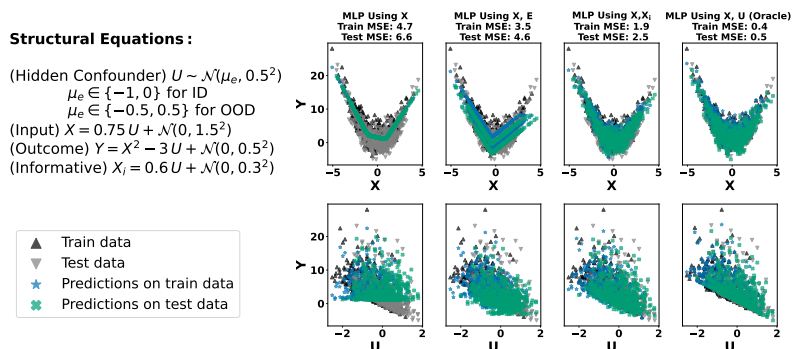


Figure D19: MLP under the high-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). Performance is strongest under high overlap, as expected.

recoverable in synthetic *linear* data whenever the observed covariates carry sufficient information about the hidden confounders, we now extend the analysis to *nonlinear* data with MLP and XGB models under both low and high confounder overlap (Figures D18–D21). Under low overlap (distant μ_e between ID and OOD data), a model trained on X alone fits an inadequate global function that cannot separate environments; incorporating environment-specific summary statistics of the observed covariates recovers the environment-specific relationships and improves performance, but the largest gains arrive only with additional informative covariates—again underscoring their central role. For completeness we report oracle results and U – Y scatter plots that make the learned input–output mappings explicit. As expected, performance is stronger under high overlap; the more consequential finding is that the method remains effective in the low-overlap regime—a previously unstudied setting that directly challenges the common-support assumption prevailing in the literature.

Effect of the causal-to-anti-causal ratio. Recent benchmarks [24] report that baseline methods (e.g. XGB, MLP) consistently match or exceed dedicated OOD generalization techniques (e.g. GDRO, IRM, VREX) on real-world tabular data, a regularity that coincides with the prevalence of the causal structure $\mathbf{X} \rightarrow Y$ in such data. To test directly whether the baselines’ advantage originates in this structure, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (so $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The baselines prove resilient even under the mixed structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ (Figure D22), attaining superior PMA-OOD scores [24] (fraction of maximum OOD accuracy), although—consistent with our theory that no single fixed strategy is universally optimal—no method dominates across all ρ (Figures D23, D24). The comparison spans a comprehensive method suite: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17],

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

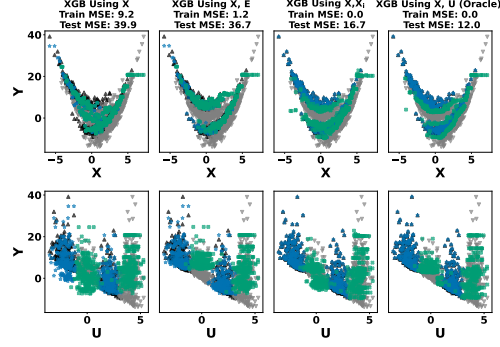


Figure D20: XGB under the low-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). As with MLP, informative covariates drive the largest improvement in the low-overlap regime.

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

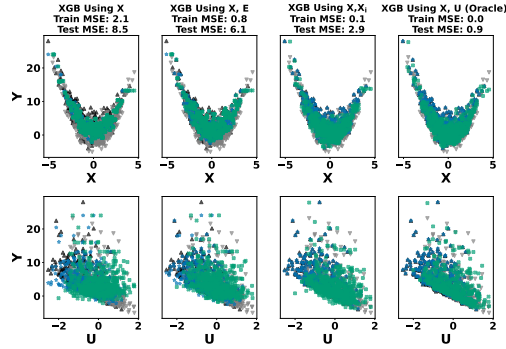


Figure D21: XGB under the high-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

888 DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) robust-
889 optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_u|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

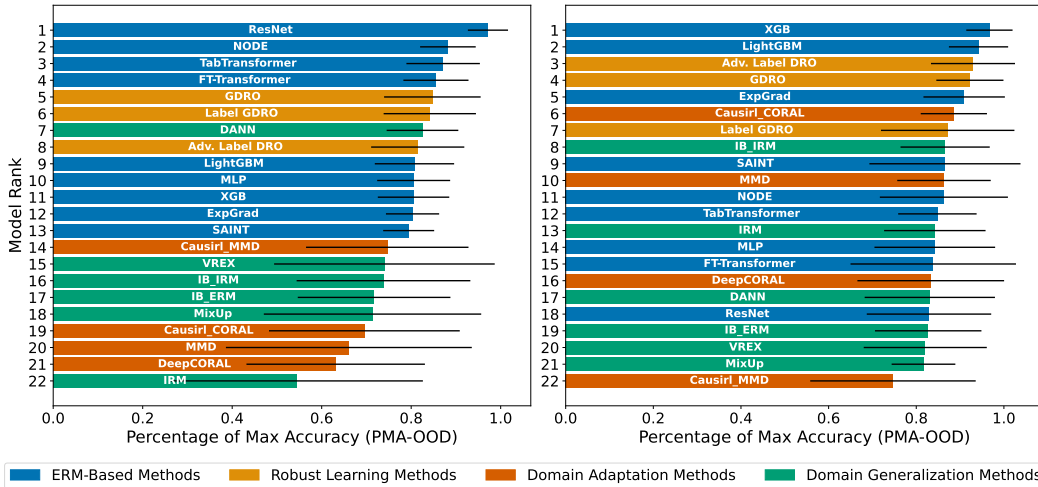


Figure D22: On the synthetic datasets, ERM-based methods attain a higher percentage of maximum OOD (PMA-OOD) test accuracy across the causal-to-anti-causal ratios, reproducing the real-world ranking of [24].

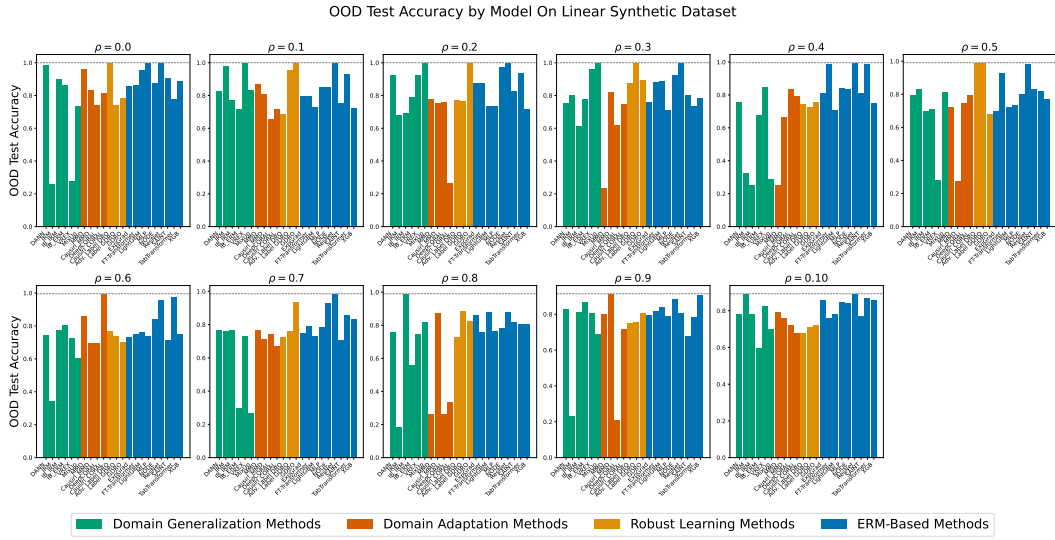


Figure D23: Per-model OOD test accuracy on the linear setting as a function of the causal-to-anti-causal ratio ρ .

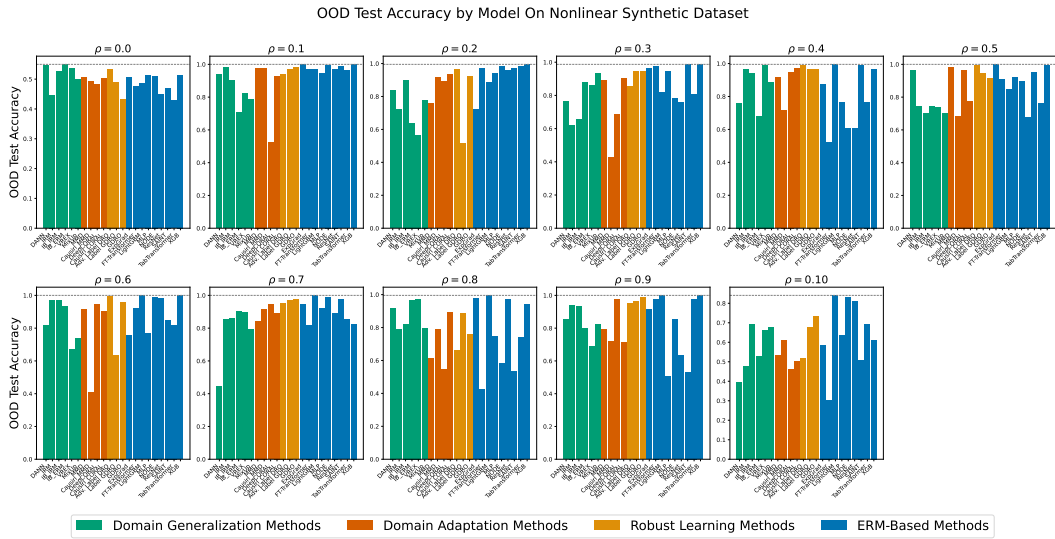


Figure D24: Per-model OOD test accuracy on the nonlinear setting as a function of the causal-to-anti-causal ratio ρ .

E Qualitative analysis of hidden confounding shift in real-world datasets

This section provides a qualitative account of hidden confounding shift in real-world datasets that complements the quantitative evidence of Table 2. Tables E3–E18 enumerate plausible hidden confounders acting on the observed covariates in a range of real-world datasets [57]; further dataset details are given in [24, 57]. We generate these candidate confounders with GPT-4o [1], and note explicitly that the resulting lists are illustrative rather than exhaustive. Taken together, they substantiate the existence and cross-domain impact of hidden confounding shift in real-world data.

Prompt. The candidate hidden confounders were elicited by querying GPT-4o with the following prompt.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.